

Abutalib-Barish Namazov

Email: hello@barish.me · Website: barish.me · LinkedIn: [abutalib-namazov](https://www.linkedin.com/in/abutalib-namazov) · Github: [BarishNamazov](https://github.com/BarishNamazov)

Education

Massachusetts Institute of Technology

Cambridge, MA

Bachelor of Science in Computer Science and Engineering; GPA: 5.0/5.0

May 2024

Leadership: Founding President of [Open-Code @ MIT](#), Founding Vice-President of [Video Gamers Alliance](#)

Experience

Software Design Group, MIT

Cambridge, MA

Software Researcher

September 2023 - May 2024

- Created a [backend framework in TypeScript](#) to support development of concept-driven, fully modular web applications.
- Designed a [HTML superset, natively reactive frontend DSL](#) to use with LLM integration. Coded interpreter in Vue 3.
- Developed [Kodless](#) from scratch using Nuxt, enabling fullstack web app creation via natural language prompts.
- Reduced student development time by over 50% in MIT classes; Kodless featured in [TEDxBoston](#) and [Forbes-sponsored MIT CSAIL talk](#).

Citadel LLC

Chicago, IL

Quantitative Research Engineering Intern in GQS

June 2023 - August 2023

- Built an end-to-end Big Data pipeline (C++ with Python bindings) to collect usage data from firm-wide APIs, managing transfer to cloud storage. Developed an analytics framework and visualization dashboard to monitor API performance and usage.
- Optimized KDB Reader by implementing efficient sorting techniques, enhancing processing speed, and conserving memory for trade and quote data extraction. Achieved 2 to 10 times speedup in specific jobs.

Microsoft

Redmond, WA

Software Engineering Intern

May 2022 - August 2022

- Developed a tool (C# and C++) to regenerate digitally signed XML licenses with custom modifications, improving debugging and enabling abstraction of licenses into readable objects.
- Enhanced license validation and C++ diagnostic tool for user-friendly output and key-holder details.

Massachusetts Institute of Technology

Cambridge, MA

Research Assistant, Teaching Assistant

February 2021 - May 2024

- Laboratory for Nuclear Science: Developed responsive web controls with ReactJS for the IsoDAR experiment, optimized Flask and C++ code for reading speeds (30-75 Hz), and [published work in NIMA](#).
- Haystack Group @ CSAIL: Contributed to a domain-specific app and research, facilitating data calculations and graph visualization for micro-journaling.
- Teaching: Helped to teach more than 2000 students over multiple semesters in classes Fundamentals of Programming, Software Construction, Web Design, and Software Design. Authored educational programming content in all of these classes.

Projects and Voluntary Work

- Taught free and public competitive-programming classes to over 8000 students in Azerbaijan in winters of [2023](#) and [2024](#).
- Co-founded [Fuiz](#), a Kahoot-like platform for online live quizzes. Coded in Rust, SvelteKit, and serverless over Cloudflare.
- Hosted a privacy and security talk in [OpenCode @ MIT](#) to help internet users understand and protect their privacy better.

Programming skills

Languages: C++, Python, Rust, JavaScript, TypeScript, HTML, CSS, Scheme, Lua

Technologies: UNIX/Bash/RHEL, CMake/Bazel, Frontend/Backend, PyBind/NumPy/Pandas/Xarray, TCP/UDP/Multicast, WebSockets,

NoSQL/SQL, Serverless/Edge/Lambda, DevOps/Deployment, CI/CD

Competitive Programming: Strong algorithmic thinking and broad knowledge of data structures; master rank in Codeforces, 6 stars in CodeChef, and O(1) rank in HackerRank (99.99th percentile)

Awards

International Olympiad in Informatics Bronze medal (2020)

International Zhautykov Olympiad Gold medal (2020)

International Olympiad of Metropolises Silver medal (2019)

National Olympiad in Informatics 3 Gold medals (2017-2019)

Junior Balkan Mathematics Olympiad Silver medal (2018)

National Olympiad in Mathematics Silver medal (2018)